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Summary

ML/AI researcher and engineer in industrial R&D, with background spanning speech recognition, quantitative research, trading and machine learning systems. Experience building research platforms and production systems across forecasting, optimisation, and large-scale data-driven modelling.

Now: working on train-time and test-time reasoning in LLM systems, applying ideas from structured search and hypothesis management developed in ASR lattice decoding. Interested in agentic workflows, programmatic prompting (DSPy), building higher-level computational systems. Where Socratic dialogue becomes an executable specification, treating RLM inference as compute, context as working memory, with explicit mechanisms for branching, verification, merging, stopping, state management.

Prior: quantitative researcher, trader and portfolio manager developing systematic equity and FX models including low SNR Signal extraction, fitting, forecasting, portfolio optimisation, trading & risk management.

Earlier: PhD research in robust speech recognition in noise, MSc in text-to-speech synthesis and spoken document retrieval. Background in applied mathematics, statistics, computer science, and industrial R&D.

Skills

Programming: C/C++/OpenMP, Python, SQL, duckdb, bash, awk, make, gdb/ddd, MATLAB, C#, R, Java.
Agents: omp/Hermes/pi/OpenCode/Codex/Claude/Cursor; local LLM workflows llama.cpp/ds4/mlx/hipfire.
Tools: Vim, Git, Github, tmux, VSCode, CLion, Jupyter, Spyder, Bloomberg & Reuters API-s.
Platforms: Linux (Ubuntu, CentOS), MacOS, MS-Windows, Gcloud, Slurm, HTCondor, Unix.

Experience

F9 Research, Director (2016–present; Harpenden, UK; remote distributed US, UK, EU)

Rekindled ML/AI interests with OSS llama.cpp, open weights local LLMs (Nemotron, DeepSeek, Qwen, GLM), agents - coding OMP/pi/Codex/OpenCode/Claude, general Hermes, data text2sql.
DNNs for tabular data forecasting (c.f. Hugging Face TabArena) using TabM, TabPFN for regression.
LLM API-s for earnings calls doc2vec features extraction for modelling and forecasting.
Quant research & development short-horizons strategies using Python, C++, cluster & cloud resources.
Managed traded a market-neutral book gross ~$350M, trade ~$35M daily, EU & US markets (MATLAB).

FutureSearch, Research Scientist (2025; remote distributed US, UK, EU)

Created then used agents to gather and organise financial data for end-user would-be products, and for internal use. Consulted on using the presumed ‘alpha’ 'generated' by the AI agent(s) for potential investment.

Marshall Wace, Senior Quantitative Researcher (2010–2016; London, UK)

Developed and scaled market-neutral portfolios from $100M to $10B+ over a period of 6 years. Pioneered wrote unified R&D framework for data ingestion, signal extraction, modelling, portfolio optimization, simulation. Mentored junior researchers, implemented reproducible research workflows.

Credit Suisse, Quantitative Analyst (2007–2009; London, UK)

Independently traded ~35M gross equity market-neutral portfolios systematically, achieving 18% lifetime returns with Sharpe 3.1. Built and operated a complete trading platform for multi-market European equities.

G-Research (DPFMG), Quantitative Analyst (2004–2007, London, UK)

Designed and implemented systematic trading models for global equities and FX, contributed to fund profitability. Modelling, forecasting, risk management and multi-period optimization for mid- and high-frequency trading strategies. Operational portfolio management and production monitoring, on-call duty.

Canon Research Europe, Researcher (2001-2004, Bracknell, UK)

Embedded Automatic Speech Recognition, indexing, and retrieval of spoken documents with speech.

Education

Ph.D. Computer Science – University of Sheffield, UK (2000)
Thesis: Robust Speech Recognition with Missing and Unreliable Data

M.Phil. Electrical Engineering – University Sv. Kiril i Metodij, Skopje, MK (1997)
Thesis: System for text-to-speech conversion for Macedonian language

B.S. Electrical Engineering – University Sv. Kiril i Metodij, Skopje, MK (1993)